

Solution Architecture

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

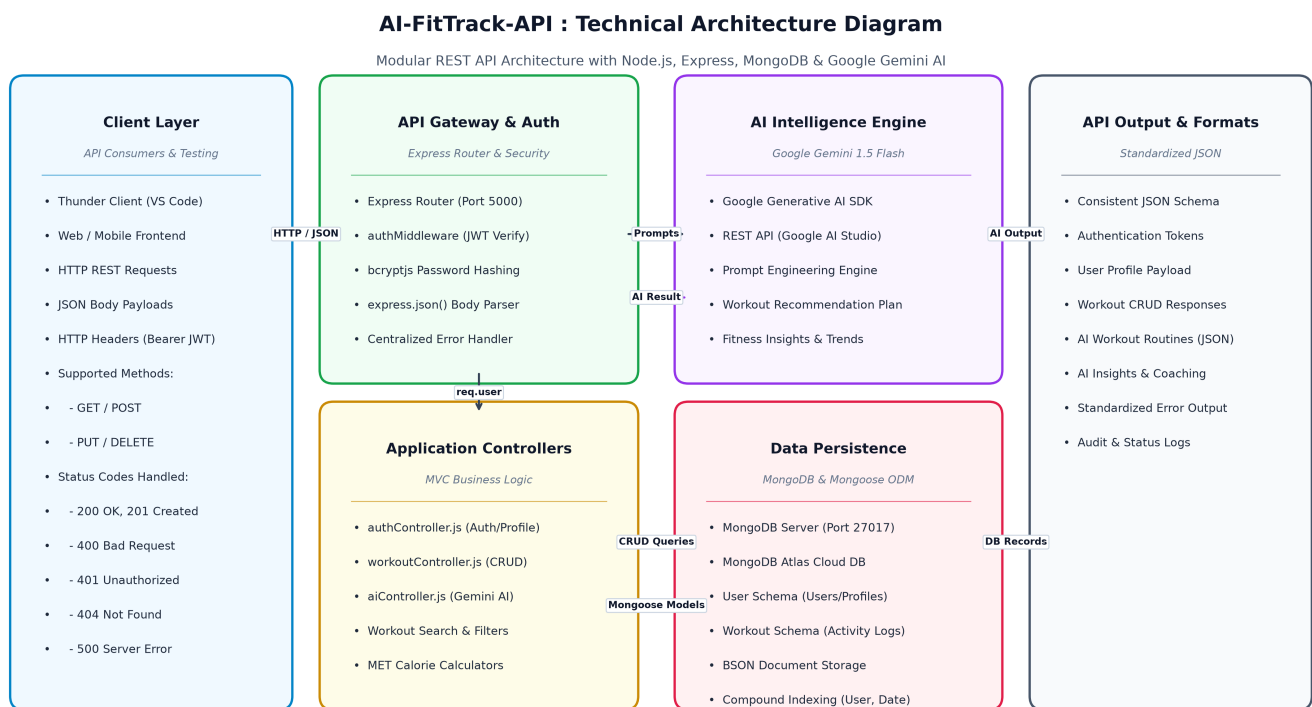


Figure 1: Modular Technical Architecture Diagram of AI-FitTrack-API

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Issues HTTP/HTTPS REST requests with JSON payloads and Bearer JWT tokens; inspects live response headers and JSON bodies.	VS Code Thunder Client, Web & Mobile Clients
API Gateway & Auth	Routes incoming traffic, extracts Bearer JWT tokens, validates signatures, attaches req.user context, and protects secure endpoints.	Express Router, authMiddleware.js, bcryptjs
Application Controllers	Executes business logic for user management, workout CRUD operations, query filtering, and external AI service calls.	authController.js, workoutController.js, aiController.js

Component Name	Description / Role in Architecture	Technologies Used
External AI Engine	Processes structured prompts containing user fitness level, goals, and equipment; returns structured JSON workout routines and insights.	Google Gemini 1.5 Flash (@google/genai SDK)
Database & Storage Layer	Persists user accounts, hashed credentials, and workout session documents with compound indexing on userId and date.	MongoDB (Port 27017 / Atlas), Mongoose ODM